

A linear-combination identity for per-subset Pillar 1: reducing strong operator vanishing to a single tail statement

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28 May 2026 (afternoon prove session)

Abstract

For λ in the Pillar 1 regime of the multi-row meta-theorem, the per-subset chain operator $M_T(S_T) := \prod_{k \in \text{tail}} A_k(S_T)$ (with $A_k = P_q^{(i_k)}$ for $k \notin S_T$ and $A_k = P_{-1}^{(i_k)}$ otherwise) is conjecturally contained in $\bigcap_{j=1}^{\tau-|S_T|} E_j^-|_{V^\lambda}$ whenever $|S_T| < \tau$ (*refined per-subset Pillar 1*). We give a linear-combination identity expressing $M_T(S_T)$ as a signed sum of *factor-removed tail chains* $\Omega_{\text{tail}}^{(\lambda)[\widehat{U}]}$ over $U \subseteq S_T$. This reduces the per-subset conjecture to a uniform *strong* statement: $\Omega_{\text{tail}}^{(\lambda)[\widehat{U}]} V^\lambda \subseteq E_1^-|_{V^\lambda}$ for every $U \subseteq \text{tail}$ with $|U| < \tau$. We prove the strong statement structurally for two distinguished tail positions of $\lambda = (2, 2, 1, 1, 1)$ via the S_{n-1} -branching method, and verify it computationally for all 11 single-factor removals on this shape. The corresponding cases of refined per-subset Pillar 1 at $|S_T| = 1$ then follow as corollaries of the identity. Remaining gaps are stated precisely: 9 tail positions whose structural proof requires arguments beyond commutation (the braid relation $S_i S_{i+1} S_i = S_{i+1} S_i S_{i+1}$ fails since $S_i S_{i+1} S_i - S_{i+1} S_i S_{i+1} = q(T_i - T_{i+1})$); the 3 outlier subsets at $|S_T| = \tau$; and the “ $\binom{\ell}{2} - 1 \in S$ ” sub-case of prefix-kill where refined per-subset Pillar 1 is needed at strength ≥ 2 .

1 Setup and statement

We use the conventions of [?], briefly recalled: $H_q(S_n)$ is the Iwahori–Hecke algebra with $T_i^2 = (q-1)T_i + q$ and braid relations; $S_i := T_i + 1$ satisfies $S_i^2 = (q+1)S_i$, so S_i has eigenvalues $0, q+1$. Set $E_i^+ := \ker(T_i - q)|_{V^\lambda} = \text{Im}(S_i|_{V^\lambda})$ and $E_i^- := \ker(S_i|_{V^\lambda})$. For $k \geq 1$, write $R'_k := S_{k-1} S_{k-2} \cdots S_1$. The *staircase chain* for S_n is $\Omega := R'_2 R'_3 \cdots R'_n$, the reduced word for $w_0 \in S_n$; total length $N = \binom{n}{2}$.

For $\lambda = (\widehat{\lambda}, 1^q) \vdash n$ with $\ell(\widehat{\lambda}) \geq 2$, every row $\widehat{\lambda}_r \geq 2$, and $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$, set $\ell := \ell(\widehat{\lambda}) + q$ and $\tau := q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|$. Split $\Omega = R \cdot \Omega_{\text{tail}}^{(\lambda)}$ with

$$R := R'_2 R'_3 \cdots R'_\ell, \quad \Omega_{\text{tail}}^{(\lambda)} := R'_{\ell+1} R'_{\ell+2} \cdots R'_n.$$

The **Pillar 1 meta-theorem** [?] states $\Omega_{\text{tail}}^{(\lambda)} V^\lambda \subseteq \bigcap_{j=1}^\tau E_j^-|_{V^\lambda}$.

Per-subset projectors and per-subset chain. $P_q^{(i)} := S_i/(q+1)$ projects onto E_i^+ ; $P_{-1}^{(i)} := I - P_q^{(i)} = (qI - T_i)/(q+1)$ projects onto E_i^- . For a subset $S \subseteq \{0, \dots, N-1\}$ of staircase positions, define

$$M(S) := \prod_{k=0}^{N-1} A_k(S), \quad A_k(S) = \begin{cases} P_{-1}^{(i_k)} & k \in S, \\ P_q^{(i_k)} & k \notin S. \end{cases}$$

Split $S = S_R \sqcup S_T$ with $S_R \subseteq \{0, \dots, \binom{\ell}{2} - 1\}$ (prefix) and $S_T \subseteq \{\binom{\ell}{2}, \dots, N - 1\}$ (tail); set

$$M_T(S_T) := \prod_{k \in \text{tail}} A_k(S_T), \quad M_R(S_R) := \prod_{k \in \text{prefix}} A_k(S_R).$$

Then $M(S) = M_R(S_R) \cdot M_T(S_T)$ and $M_T(\emptyset) = \Omega_{\text{tail}}^{(\lambda)} / (q+1)^{|\text{tail}|}$.

Factor-removed tail chains. For $U \subseteq \text{tail}$, let $\Omega_{\text{tail}}^{(\lambda)}[\widehat{U}] \in H_q$ denote the product of S -factors at the tail positions *not in* U :

$$\Omega_{\text{tail}}^{(\lambda)}[\widehat{U}] := \prod_{k \in \text{tail} \setminus U} S_{i_k},$$

preserving the left-to-right staircase order. Thus $\Omega_{\text{tail}}^{(\lambda)}[\widehat{\emptyset}] = \Omega_{\text{tail}}^{(\lambda)}$ and $\Omega_{\text{tail}}^{(\lambda)}[\widehat{U}]$ has $|\text{tail}| - |U|$ factors.

2 The linear-combination identity

Proposition 1 (Per-subset M_T as a signed sum of $\Omega_{\text{tail}}^{(\lambda)}[\widehat{U}]$). *For any $S_T \subseteq \text{tail}$,*

$$(q+1)^{|\text{tail}|} M_T(S_T) = \sum_{U \subseteq S_T} (-1)^{|S_T| - |U|} (q+1)^{|U|} \Omega_{\text{tail}}^{(\lambda)}[\widehat{U}]. \quad (1)$$

Proof. For $k \in S_T$, $A_k = I - P_q^{(i_k)} = I - S_{i_k} / (q+1)$, so we write

$$A_k = \frac{1}{q+1} ((q+1)I - S_{i_k}), \quad k \in S_T,$$

while for $k \notin S_T$, $A_k = S_{i_k} / (q+1)$. Therefore

$$M_T(S_T) = \prod_k A_k(S_T) = \frac{1}{(q+1)^{|\text{tail}|}} \prod_{k \in \text{tail}} \tilde{A}_k,$$

where $\tilde{A}_k = S_{i_k}$ for $k \notin S_T$ and $\tilde{A}_k = (q+1)I - S_{i_k}$ for $k \in S_T$. Expand each $(q+1)I - S_{i_k}$ as a sum over a binary choice $c_k \in \{0, 1\}$: $c_k = 0$ contributes $(q+1)I$ at position k ; $c_k = 1$ contributes $-S_{i_k}$. Distributing the product (in the fixed staircase order) gives

$$\prod_{k \in \text{tail}} \tilde{A}_k = \sum_{V \subseteq S_T} (-1)^{|V|} (q+1)^{|S_T| - |V|} \prod_k [I \text{ for } k \in S_T \setminus V, S_{i_k} \text{ else}].$$

The product on the right is $\Omega_{\text{tail}}^{(\lambda)}[\widehat{(S_T \setminus V)}]$: we have I at the $|S_T \setminus V|$ positions in $S_T \setminus V$ (which deletes those factors) and S_{i_k} at every other position. Substituting $U := S_T \setminus V$:

$$\prod_k \tilde{A}_k = \sum_{U \subseteq S_T} (-1)^{|S_T| - |U|} (q+1)^{|U|} \Omega_{\text{tail}}^{(\lambda)}[\widehat{U}],$$

and dividing by $(q+1)^{|\text{tail}|}$ yields (??). □

$$[|S_T| = 0, 1, 2]$$

$$(q+1)^{|\text{tail}|} M_T(\emptyset) = \Omega_{\text{tail}}^{(\lambda)},$$

$$(q+1)^{|\text{tail}|} M_T(\{p\}) = (q+1) \Omega_{\text{tail}}^{(\lambda)}[\widehat{\{p\}}] - \Omega_{\text{tail}}^{(\lambda)},$$

$$(q+1)^{|\text{tail}|} M_T(\{p, q\}) = (q+1)^2 \Omega_{\text{tail}}^{(\lambda)}[\widehat{\{p, q\}}] - (q+1) \Omega_{\text{tail}}^{(\lambda)}[\widehat{\{p\}}] - (q+1) \Omega_{\text{tail}}^{(\lambda)}[\widehat{\{q\}}] + \Omega_{\text{tail}}^{(\lambda)}.$$

3 Strong Pillar 1 reduction

Definition 2 (Strong Pillar 1). Let λ be in the Pillar 1 regime. The *Strong Pillar 1 property at level s* is the statement

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{U}]} V^\lambda \subseteq E_1^-|_{V^\lambda} \quad \forall U \subseteq \text{tail with } |U| \leq s. \quad (2)$$

For $s = 0$, this is Pillar 1 (whose conclusion is in fact stronger: $\subseteq \bigcap_{j=1}^\tau E_j^-$). For $s \geq 1$ it is a nontrivial extension.

Theorem 3 (Per-subset Pillar 1 from Strong Pillar 1). *If the Strong Pillar 1 property at level s holds (Definition ??), then for every $S_T \subseteq \text{tail}$ with $|S_T| \leq s$,*

$$M_T(S_T) V^\lambda \subseteq E_1^-|_{V^\lambda}.$$

Proof. By Proposition ??,

$$(q+1)^{|\text{tail}|} M_T(S_T) V^\lambda \subseteq \sum_{U \subseteq S_T} \Omega_{\text{tail}}^{(\lambda)[\widehat{U}]} V^\lambda \subseteq E_1^-$$

where the last inclusion uses the Strong Pillar 1 hypothesis for every $U \subseteq S_T$, each of size $\leq s$. □

Remark 4 (Refined per-subset Pillar 1). The stronger refinement $M_T(S_T) V^\lambda \subseteq \bigcap_{j=1}^{\tau-|S_T|} E_j^-$ (Conjecture 3 of [?]) likewise reduces to a *refined Strong Pillar 1*: $\Omega_{\text{tail}}^{(\lambda)[\widehat{U}]} V^\lambda \subseteq \bigcap_{j=1}^{\tau-|U|} E_j^-$ for every $U \subseteq \text{tail}$ with $|U| < \tau$, by the same proof. At $\lambda = (2, 2, 1, 1, 1)$ ($\tau = 2$) this gives, at $|U| = 1$, the bound $\Omega_{\text{tail}}^{(\lambda)[\widehat{U}]} V^\lambda \subseteq E_1^-$, matching Definition ??.

4 Strong Pillar 1 at $\lambda = (2, 2, 1, 1, 1)$: two structural proofs

We focus on $\lambda = (2, 2, 1, 1, 1)$ where $\widehat{\lambda} = (2, 2)$, $\ell(\widehat{\lambda}) = 2$, $q = 3$, $\ell = 5$, $\tau = 2$, $n = 7$, $N = 21$. The tail consists of positions 10 through 20 corresponding to staircase letters $(5, 4, 3, 2, 1; 6, 5, 4, 3, 2, 1)$: blocks $R'_6 = S_5 S_4 S_3 S_2 S_1$ (positions 10–14) and $R'_7 = S_6 S_5 S_4 S_3 S_2 S_1$ (positions 15–20).

The $S_{n-1} = S_6$ branching of V^λ is

$$V^\lambda \downarrow S_6 \cong V^{(2,1,1,1,1)} \oplus V^{(2,2,1,1)},$$

where we denote the summands W_1, W_2 respectively.

Lemma 5 (Structural Strong Pillar 1 at position 10). $\Omega_{\text{tail}}^{(\lambda)[\widehat{\{10\}}]} V^\lambda \subseteq E_1^-|_{V^\lambda}$.

Proof. Position 10 corresponds to the leftmost S_5 of R'_6 . Removing it leaves $R'_6[\widehat{S_5}] = S_4 S_3 S_2 S_1 = R'_5$, so

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{\{10\}}]} = R'_5 \cdot R'_7 = R'_5 \cdot (S_6 R'_6),$$

using $R'_7 = S_6 R'_6$.

Step 1: Commutation. The letters of $R'_5 = S_4 S_3 S_2 S_1$ are all ≤ 4 , so $|6 - i| \geq 2$ for each i and S_6 commutes with every factor of R'_5 . Therefore $R'_5 \cdot S_6 = S_6 \cdot R'_5$, giving

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{\{10\}}]} = S_6 \cdot (R'_5 R'_6).$$

Step 2: $R'_5 R'_6$ acts via S_6 -branching. Since $R'_5 R'_6 \in H_q(S_6)$ (no T_6), the operator preserves the $W_1 \oplus W_2$ decomposition.

On $W_2 = V^{(2,2,1,1)}$: the shape $\nu = (2, 2, 1, 1)$ has $\widehat{\nu} = (2, 2)$, $\ell(\widehat{\lambda}) = 2$, $q'' = 2$, satisfying the Pillar 1 hypotheses ($q'' \geq |\widehat{\nu}| - 2\ell(\widehat{\lambda}) + 2 = 2$, $\tau(\widehat{\nu}) = 1$). The chain $R'_5 R'_6 \in H_q(S_6)$ is exactly $\Omega_{\text{tail}}^{(\lambda)(2,2,1,1)}$ (blocks at indices $\ell(\nu) + 1 = 5$ and $\ell(\nu) + 2 = 6$), so Pillar 1 applies:

$$R'_5 R'_6 W_2 \subseteq E_1^-|_{W_2}.$$

On $W_1 = V^{(2,1,1,1)}$: this is a hook with $k_h = 2$, $m_h = 4$, satisfying $m_h \geq k_h$. The hook column-1 theorem [?, Thm. 6] gives $R'_6 W_1 \subseteq E_1^-|_{W_1}$. Then $R'_5 = S_4 S_3 S_2 S_1$ has rightmost factor S_1 , which is identically zero on $E_1^- = \ker(S_1)$:

$$R'_5 \cdot R'_6 W_1 = S_4 S_3 S_2 \cdot (S_1 \cdot E_1^-|_{W_1}) = 0.$$

Combining, $R'_5 R'_6 V^\lambda = R'_5 R'_6(W_1) \oplus R'_5 R'_6(W_2) \subseteq 0 \oplus E_1^-|_{W_2} \subseteq E_1^-|_{V^\lambda}$.

Step 3: S_6 preserves E_1^- . Since $|6 - 1| = 5 \geq 2$, S_6 commutes with T_1 and therefore preserves E_1^- . Hence

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{10}]} V^\lambda = S_6 \cdot R'_5 R'_6 V^\lambda \subseteq S_6 E_1^- \subseteq E_1^- . \quad \square$$

□

Lemma 6 (Structural Strong Pillar 1 at position 15). $\Omega_{\text{tail}}^{(\lambda)[\widehat{15}]} V^\lambda \subseteq E_1^-|_{V^\lambda}$.

Proof. Position 15 corresponds to the leftmost S_6 of R'_7 . Removing it leaves $R'_7[\widehat{S_6}] = S_5 S_4 S_3 S_2 S_1 = R'_6$, so

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{15}]} = R'_6 \cdot R'_6 = (R'_6)^2.$$

Step 1: Algebraic identity. Using $R'_6 = S_5 \cdot R'_5$ ($R'_6 = S_5 S_4 S_3 S_2 S_1$), associativity gives

$$(R'_6)^2 = (S_5 R'_5) \cdot R'_6 = S_5 \cdot R'_5 R'_6.$$

Step 2: $R'_5 R'_6 \in H_q(S_6)$ and the same branching argument as Lemma ?? gives $R'_5 R'_6 V^\lambda \subseteq E_1^-|_{V^\lambda}$.

Step 3: S_5 preserves E_1^- . Since $|5 - 1| = 4 \geq 2$, S_5 commutes with T_1 and therefore preserves E_1^- . Hence

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{15}]} V^\lambda = S_5 \cdot R'_5 R'_6 V^\lambda \subseteq S_5 E_1^- \subseteq E_1^- . \quad \square$$

□

Corollary 7 (Per-subset Pillar 1 at $|S_T| = 1$ for positions 10, 15). *For $S_T \in \{\{10\}, \{15\}\}$,*

$$M_T(S_T) V^\lambda \subseteq E_1^-|_{V^\lambda}.$$

Proof. By Theorem ?? and Lemmas ??–??. □

□

Table 1: $\Omega_{\text{tail}}^{(\lambda)}[\widehat{p}]V^\lambda$ for each tail position p on $\lambda = (2, 2, 1, 1, 1)$. Rank, E_1^- and E_2^- containment verified by exact computation.

position p	letter i_p	block	rank of $\Omega_{\text{tail}}^{(\lambda)}[\widehat{p}] _{V^\lambda}$	$\subseteq E_1^-$	$\subseteq E_2^-$	structural proof?
10	5	R'_6 (leftmost)	1	yes	yes	Lemma ??
11	4	R'_6	1	yes	no	empirical only
12	3	R'_6	0	yes	yes	empirical only
13	2	R'_6	0	yes	yes	empirical only
14	1	R'_6 (rightmost)	1	yes	yes	empirical only
15	6	R'_7 (leftmost)	1	yes	no	Lemma ??
16	5	R'_7	0	yes	yes	empirical only
17	4	R'_7	0	yes	yes	empirical only
18	3	R'_7	1	yes	yes	empirical only
19	2	R'_7	1	yes	yes	empirical only
20	1	R'_7 (rightmost)	1	yes	yes	empirical only

5 Computational verification of Strong Pillar 1

We verify the full Strong Pillar 1 property at $\lambda = (2, 2, 1, 1, 1)$ for all single-factor removals and identify the residual cases at pair removals. Computations use exact rationals at $q = 5/7$, cross-checked at $q = 2/3$.

All 11 single-removals lie in E_1^- ; 9 of them also lie in E_2^- (the sharper bound). Combining with the linear-combination identity (Proposition ??), we obtain $M_T(\{p\})V^\lambda \subseteq E_1^-$ for every singleton S_T .

For pair removals $|U| = 2$, 52 of the $\binom{11}{2} = 55$ pairs satisfy $\Omega_{\text{tail}}^{(\lambda)}[\widehat{U}]V^\lambda \subseteq E_1^-$; the remaining 3 *outliers* are $U \in \{\{11, 12\}, \{12, 15\}, \{15, 16\}\}$, matching exactly the three $|S_T| = \tau$ outlier subsets identified in [?]. At $|S_T| = 2 = \tau$, refined per-subset Pillar 1 makes no claim, so the outliers are outside the scope of Theorem ?? (level $s = 1$).

6 Why braid does not help (and what does)

A natural temptation is to use the Coxeter braid relation $S_i S_{i+1} S_i = S_{i+1} S_i S_{i+1}$ to manipulate tail chains, e.g., to extend the Lemma ?? commutation argument to position 11. This fails for $S_i = T_i + 1$:

Lemma 8 (S -braid defect). *In H_q , $S_i S_{i+1} S_i - S_{i+1} S_i S_{i+1} = q(T_i - T_{i+1})$.*

Proof. Expanding $(T_i + 1)(T_{i+1} + 1)(T_i + 1)$ and using $T_i^2 = (q - 1)T_i + q$:

$$S_i S_{i+1} S_i = T_i T_{i+1} T_i + T_i T_{i+1} + T_{i+1} T_i + (q + 1)T_i + T_{i+1} + (q + 1).$$

Symmetrically, $S_{i+1} S_i S_{i+1} = T_{i+1} T_i T_{i+1} + T_{i+1} T_i + T_i T_{i+1} + (q + 1)T_{i+1} + T_i + (q + 1)$. Subtracting (using $T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}$) yields $(q + 1)(T_i - T_{i+1}) - (T_i - T_{i+1}) = q(T_i - T_{i+1})$. $\square$ $\square$

Remark 9 (Why positions 10 and 15 are special). The proofs of Lemmas ??–?? use only: (a) the commutation $S_i S_j = S_j S_i$ for $|i - j| \geq 2$ (Coxeter commutation); and (b) the trivial associativity

identity $S_5 R'_5 = R'_6$. No braid relation is invoked. The two privileged positions are exactly the *leftmost* factors of R'_6 and R'_7 (positions 10 and 15, letters 5 and 6), because removing the leftmost S_5 of R'_6 leaves exactly R'_5 , and removing the leftmost S_6 of R'_7 leaves exactly R'_6 . Both produce a standard $\Omega_{\text{tail}}^{(\lambda)}$ -like chain (for a smaller shape or by squaring) that succumbs to Pillar 1 plus S_1 -kill on the W_1 -summand. The remaining 9 positions yield modified chains that are not of this standard form; structural proofs would require either an enhanced form of Pillar 1 or genuine use of the S -braid defect from Lemma ?? (with the correction term integrated into the analysis).

7 Status and gaps

Established.

- Proposition ?? : linear-combination identity expressing $M_T(S_T)$ as a signed sum of factor-removed tail chains, with coefficients $\pm(q+1)^{|U|}$.
- Theorem ?? : per-subset Pillar 1 at level s reduces to Strong Pillar 1 at level s , a uniform statement depending only on $|U|$ and not on the choice of P_q vs. P_{-1} at each position.
- Lemmas ?? and ?? : Strong Pillar 1 at the distinguished positions 10 and 15 at $\lambda = (2, 2, 1, 1, 1)$, by an S_{n-1} -branching argument using only commutation and Pillar 1 on smaller shapes.
- Corollary ?? : per-subset Pillar 1 at $|S_T| = 1$ for the two singleton subsets at positions 10 and 15.
- Lemma ?? : explicit q -correction in the S -braid identity.

Open gaps (precisely stated).

- Strong Pillar 1 at positions 11, 12, 13, 14, 16, 17, 18, 19, 20 (nine positions) for $\lambda = (2, 2, 1, 1, 1)$: verified computationally; structural proof open. Of these, positions 12, 13, 16, 17 have $\Omega_{\text{tail}}^{(\lambda)}[\{p\}] = 0$ as an operator on V^λ (a strictly stronger statement than $\subseteq E_1^-$), suggesting they have their own clean argument we have not yet identified.
- Strong Pillar 1 at level $s = 2$ for 52 of the 55 pair-removals. The 3 outliers correspond to $|S_T| = \tau$ and lie outside refined per-subset Pillar 1; they require a separate inductive argument on Ω_{S_ℓ} -kernels in branching summands (the residual gap from [?]).
- The prefix-kill cases where $\binom{\ell}{2} - 1 \in S$ (position 9 in S): refined per-subset Pillar 1 at level ≥ 2 would close these.
- Strong Pillar 1 at general $\lambda = (2, 2, 1^m)$ for $m \geq 4$: computational verification beyond $m = 3$ is open, as is the inductive structure of the argument across m .

Methodological note. The two structural Lemmas use the S_{n-1} -branching method (Lemma ?? Step 2): an operator that lies in $H_q(S_{n-1})$ acts on $V^\lambda \downarrow S_{n-1}$ summand-by-summand. On each summand V^ν , smaller-shape Pillar 1 or the hook column-1 theorem applies. The crucial input is that the *leftmost* factor of the tail (the S_5 at position 10, or the S_6 at position 15) is the one whose removal yields a chain in $H_q(S_{n-1})$. For positions deeper in R'_6 or R'_7 , the resulting chain contains S_5 or S_6 entangled with the lower letters, and the $H_q(S_{n-1})$ factorisation requires a braid-like manipulation that introduces the $q(T_i - T_{i+1})$ correction of Lemma ?? . We have not found a clean way to integrate this correction into the branching argument in this session.

References

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